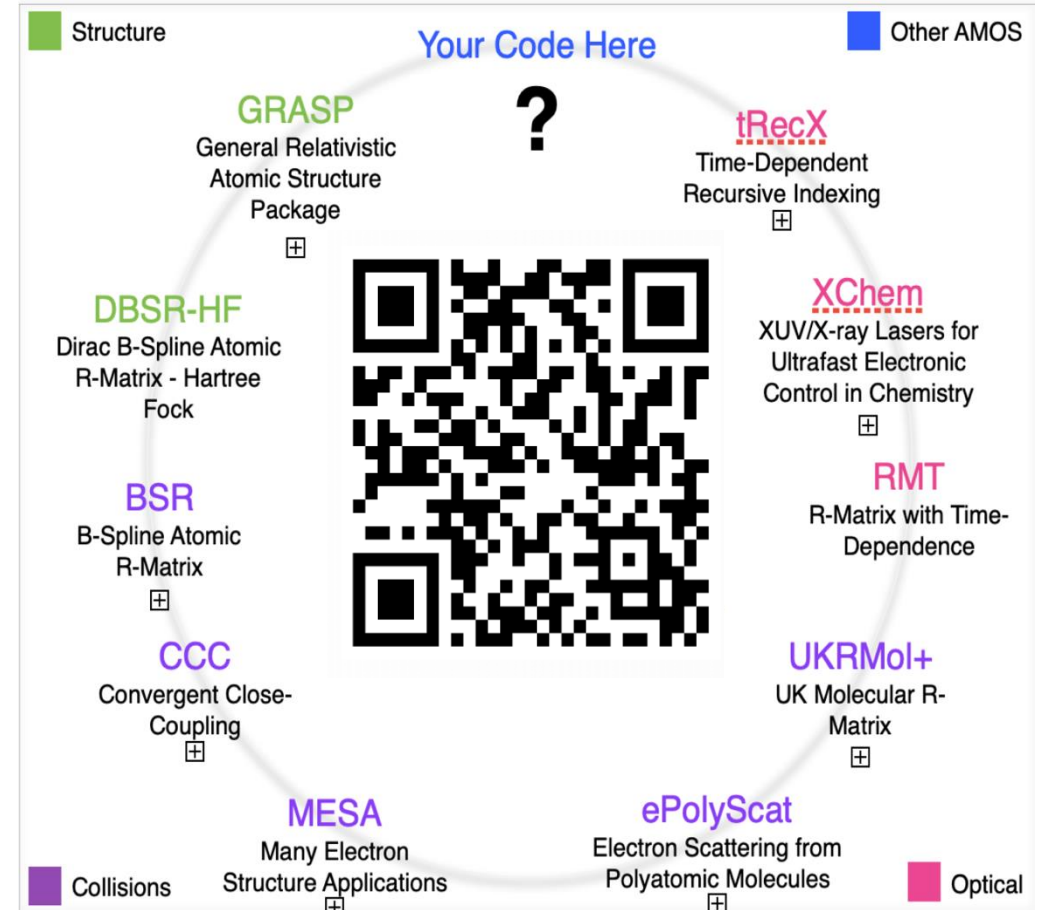
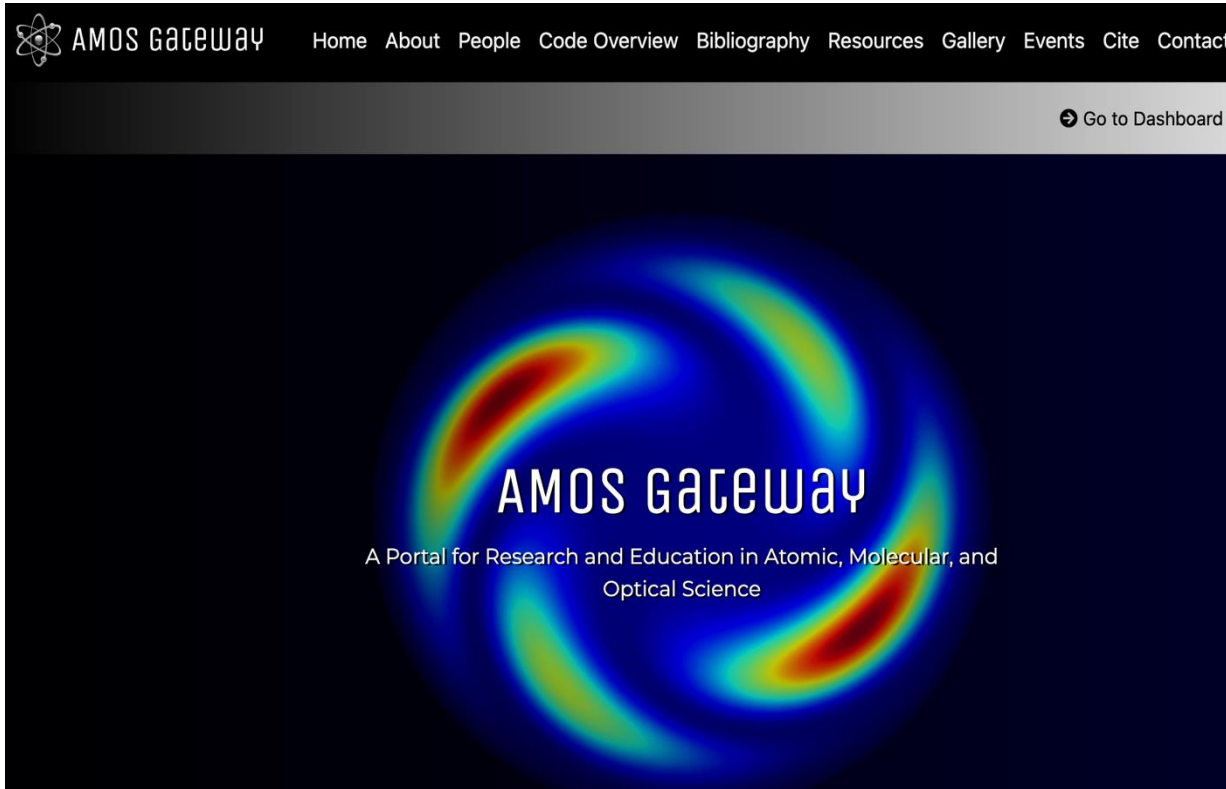


Atomic Molecular Optical Science Through HPC Simulations

Sudhakar Pamidighantam
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Atomic Molecular and Optical Sciences online



AMOS BSR Workflow with Custom Django Interfaces

Experiment status: **UNSUBMITTED** Job status: **UNSUBMITTED**

Run Type

MODULE

UTILITY

WORKFLOW

BOUND

STGF

SCATTERING

Input Files

Upload

name.bsw

bsr_par

target

knot.dat

name.c

BOUND

STGF

SCATTERING

target

Basic Data

Target States

Partial Waves

Title

e + C

Coupling Type

LS

Atomic Number

6

Number of Electrons

6

Upload

target

dstgf

H.DAT

BOUND

STGF

SCATTERING

Upload

name.bsw

bsr_par

target

knot.dat

name.c

Parameters

Index of the initial partial wave

1

Index of the final partial wave

4

Select a Compute Resource

Bridges2

✓ Expanse

Frontera

Ookami

stampede2.tacc.utexas.edu

Stampede3

Allocation

Bartschat Allocation

Settings for queue compute

1

64

1330 minutes

3000 MB

NODE COUNT

CORE COUNT

TIME LIMIT

PHYSICAL MEMORY

Continuation of Carbon Scattering

Experiment status: **COMPLETED** Job status: **COMPLETED**

Input Files

H.DAT

Plottables

Intermediate Files

TMAT_keep

Logs

Files

tr_nnn_nnn

tr_001_001

tr_001_002

tr_001_003

tr_002_002

tr_002_003

Select Plot Parameters

new

☒ Energy (eV)

☐ Cross-section

☐ Energy (Ry/z^2)

☒ Collision Strength

Flags (help)

Enter flags here

Create Plot

Plotted Files

☒ tr_001_001

☒ tr_001_002

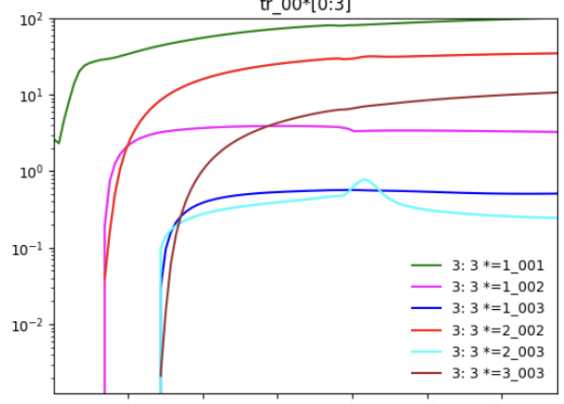
☒ tr_001_003

☒ tr_002_002

☒ tr_002_003

☒ tr_003_003

tr_00*[0:3]



Open-Source software Frameworks

<https://github.com/apache/airavata>

<https://github.com/gt-artisan/AMOS-Gateway>

<https://github.com/SciGaP/epolyscat-django-app>



CUSTOS
Secure
Authentication and
Authorization



Managed File
Transfer (MFT)



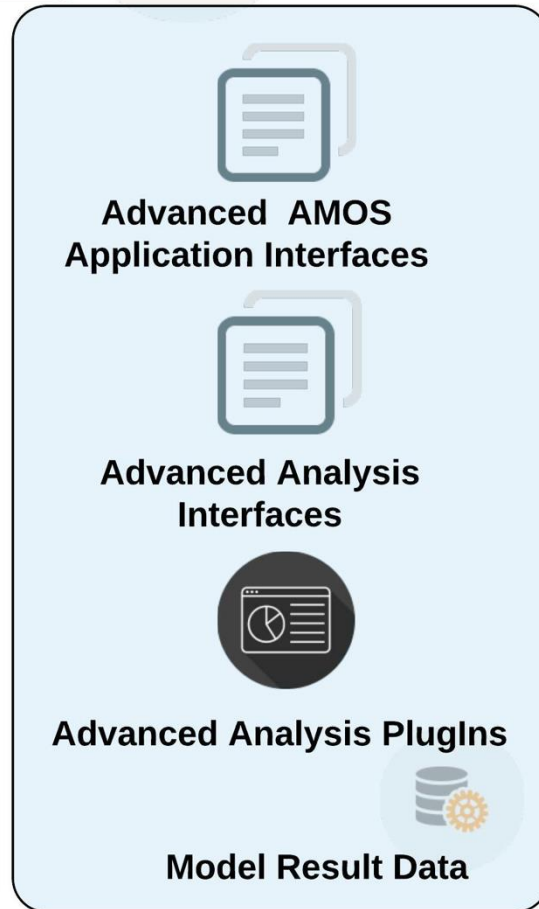
Gateway
Management



Research and
Student User
Community



AMOS Data
Management



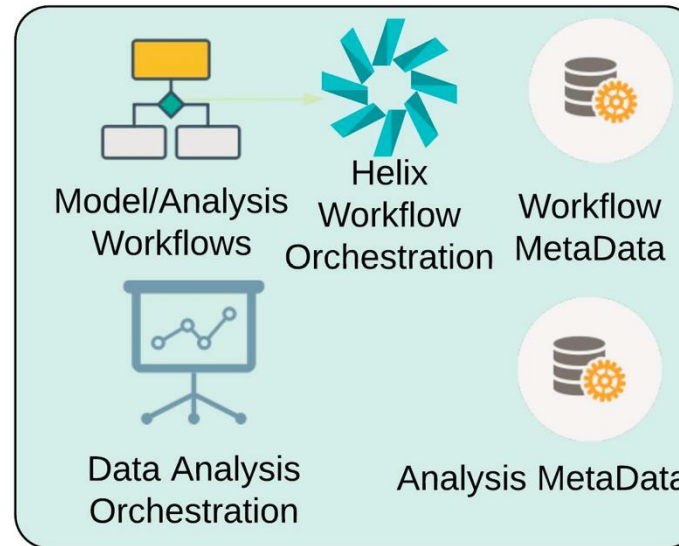
**Advanced AMOS
Application Interfaces**

**Advanced Analysis
Interfaces**

Advanced Analysis Plugins



Model Result Data



Model/Analysis
Workflows

Helix
Workflow
Orchestration

Workflow
MetaData

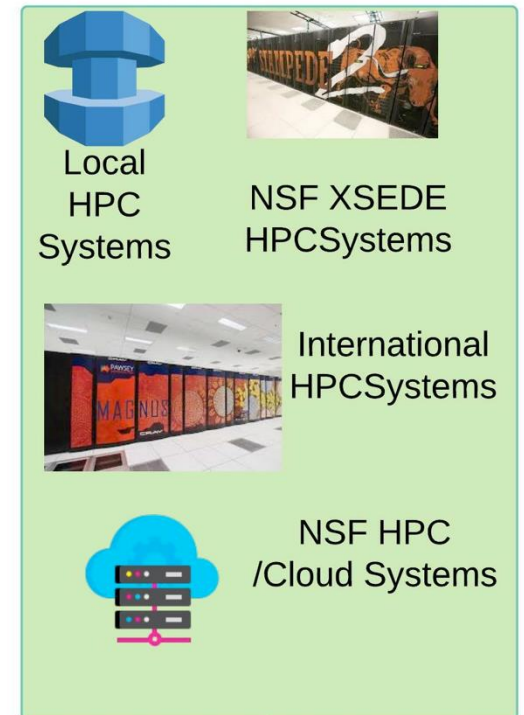
Data Analysis
Orchestration

Analysis MetaData

POWERED BY



Hosted by



Local
HPC
Systems

NSF XSEDE
HPCSystems

International
HPCSystems

NSF HPC
/Cloud Systems



Third-Party
AMOS Data
Repository



Third-Party
Tools